

This is the basis of the stereoscopic television models that are being launched now.

"3D technology works on the basis of segregating the left and right frames of any image, which results in the creation of a sense of depth," explains Tadata Kimura, general manager-marketing, Sony India.

Companies such as Panasonic, NVIDIA and Sony are betting on the use of shutter glasses to achieve this 3D experience. These glasses use lenses which are actually small liquid-crystal display (LCD) screens, and alternately shut off the right and the left eye. The television, in turn, alternately telecasts images meant for the left and right eye (stored in the even and odd fields, respectively, of its video signal). The brain processes these into a 3D image for the viewer. The TV coordinates with the shutter glasses via networking technologies like radio frequency (RF) or infrared (IR), to show the correct image corresponding to the eye that is open. The shutting of eyes happens so fast and so smoothly that the user would not even feel it. However, this effectively reduces the television's refresh time by half, and could cause headache in some users.

"In high-definition 3D technology, two HD video streams are generated by two HD cameras kept in a single body, as in Panasonic's AG-3DA camcorder, or two independent cameras kept on a rig. These two HD, left-hand/right-hand video streams are generated by 3D cameras or camera systems exactly in the same way as we see with our two eyes. Therefore in 3D television, two stereoscopic pictures/videos are handled or processed, while in 2D there is only a single video stream/picture. Therefore in 3D television there is an additional capability to handle two video streams, to show two pictures on the screen sequentially. As the 3D TV has to handle two sets of moving pictures, the picture refreshing rate has to be more than double of 2D television. Normally, 3D televisions' refresh rate is four times that of 2D

How are 3D films shot?

3D film-making requires:

1. Special cameras and rigs to shoot from two or more angles, to obtain images that can be combined to form a 3D image
2. A complex understanding of the concept of depth and how the human mind perceives it
3. An empathising of where and how the user will view the movie, and suitability of the images to his viewing positions
4. Complex editing, and in some cases 2D-to-3D conversion tools, working with high-performance computing platforms
5. High level of skills to maintain parallax, vertical gap and the like

However, brands such as Panasonic, Fujifilm and Sony now have movie cameras meant especially for 3D which reduce the burden significantly for film-makers. With the Panasonic 3D camera recorder (AG-3DA1), for instance, the entire complexity has been packaged into an easy-to-handle body that contains two lenses and automatic correction for horizontal and vertical displacement, and image convergence. It has been found that with a traditional 3D rig, one is able to shoot in a few locations in one day, while with the Panasonic 3D camera recorder, one could shoot at over 15 separate locations and cover 161 km in the same period of time.

Panasonic has also launched a consumer-level 3D camcorder (SDT750), which requires the user to add only a 3D conversion lens to automatically enable 3D shoots. Left- and right-eye images are simultaneously shot with two lenses, and the camera creates 3D images by artificially reproducing visual disparity.

Sony too is working on a prototype (EX3), which is sure to give Panasonic's model tough competition.

In India, the Prasad Group recently launched a range of latest 3D movie cameras and other equipment, in addition to a complete, end-to-end 3D movie production solution that includes the production, post-production and delivery of 3D content for cinemas and other media.

TVs, that is, 200 or 240 times per second in the case of 3D HDTV compared to 50 or 60 times of normal 2D TV," explains Sharma.

The other possibility is use of polarised glasses—an option that has been used for long by movie houses such as IMAX. The two lenses of the glass are polarised differently, thereby showing different images to each eye. However, in this case, the movie has to be shot using two cameras or a single camera with two lenses. Correspondingly, two projectors fitted with polarising filters on their lenses are required to show the movie. When viewed through the polarised glasses, the left eye sees only those images which the left projector is screening, while the viewer's right eye sees only those images which the right projector is screening. As both the images are taken from different angles, the viewer's brain combines the two to come up with a single 3D image. This technique is used more for theatre viewing, but there have been some examples of

products that use modified cathode ray tube (CRT) monitors and polarised glasses for 3D viewing.

Both techniques have their own advantages and disadvantages, but, in general, the biggest drawback is that you have to wear glasses to watch 3D video, as the picture would appear blurred and useless without the glasses. Despite the fact that today's glasses are quite fashionable and comfortable, wearing them is irritating in hot and humid weathers, let alone in households that have naughty brats who might break the expensive glasses.

3D without glasses

TV makers such as Panasonic and LG are working on auto-stereoscopic 3D televisions that would not require the viewer to wear any special glasses for 3D viewing. That would be a welcome change, when it arrives, and perhaps go a long way in increasing the popularity of 3D television.

Such televisions could use one of two technologies to create this 3D ef-